Nonlinear Stabilisation of the Electron Soliton

Documentation for the OpenWave M4 Implementation

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Abstract

This document provides a self-contained technical specification for the nonlinear wave equation that is proposed to stabilise the electron core ($K_{WC} = 10$) in the vector M4 model of the OpenWave platform. The coupling coefficient is derived from the BCC lattice geometry of Energy Wave Theory, three complementary forms of the nonlinear term are presented, and the role of the Degraded EMC Wall as a necessary boundary condition is quantified. The 1-3-6 wave-centre arrangement is shown to be the unique self-consistent candidate satisfying the topological and energetic constraints of the vacuum lattice. The stability of this configuration is a postulate whose numerical verification is the immediate objective of the M4 implementation described herein.

1 Context and Objective

The scalar M3 model of OpenWave has demonstrated that spherical phyllotaxis (golden-angle, $\sim 137.5^\circ$) resolves the K-selectivity problem at the tested neighbouring values: $K = 10$ forms a stable standing wave (amplitude flat at ~ 0.29 after ~ 140 steps under a 0.1 perturbation), while $K = 9$ and $K = 11$ decay systematically (-34% and -45%, respectively, with $K = 9$ confirmed across two independent runs).[1] In the vector M4 model, however, the same golden-angle arrangement does *not* stabilise the electron core, indicating that the missing ingredient is not purely geometric but nonlinear.

2 General Soliton Wave Equation

The scalar amplitude $\Psi(\mathbf{r}, t)$ of the standing wave is governed by

$$\Big(\frac{\partial^2}{\partial t^2} - c^2\nabla^2\Big)\Psi(\mathbf{r}, t) + \mathcal{F}(\Psi) = 0, \tag{1}$$

where c is the speed of light and $\mathcal{F}$ encodes the nonlinear self-interaction that prevents dispersive collapse.

3 Coupling Coefficient from BCC Geometry

In EWT the magnetic deficit ϵ_M is the fundamental lattice response parameter. It is defined through the calibrated stiffness N_{final} and the geometric limit $8\pi^4$:

$$\epsilon_M = \frac{1}{N_{\text{final}}\,\pi^3} \;\approx\; \frac{1}{8\pi^7}, \qquad N_{\text{final}} = 778.818123. \tag{2}$$

[1] Pull Request #205, `https://github.com/openwave-labs/openwave/pull/205`, merged; closes/contributes to Issue #203.

The approximate equality reflects the 0.058% lattice impedance δ between N_{final} and the ideal BCC packing limit $8\pi^4$. Physically, ϵ_M measures the fractional loss of stiffness caused by the soliton's spin-induced magnetic torque.

The natural coupling strength for the nonlinear term is the inverse deficit,

$$\gamma \equiv \frac{1}{\epsilon_M} = N_{\text{final}}\,\pi^3 \approx 2.4148 \times 10^4. \tag{3}$$

For numerical implementation two variants are available:

- $\gamma_{\text{geo}} = 8\pi^7 \approx 2.4162 \times 10^4$ (ideal BCC limit);
- $\gamma_{\text{final}} = N_{\text{final}}\pi^3 \approx 2.4148 \times 10^4$ (anchored to the measured α).

The 0.058% difference may be critical for discriminating $K = 10$ from neighbouring configurations. The M4 implementation should initially use γ_{final} and, if necessary, explore the geometric limit as a consistency check.

4 Three Complementary Forms of $\mathcal{F}$

Variant A — Pure cubic nonlinearity

The simplest form compatible with NLS-type soliton equations is

$$\boxed{\mathcal{F}(\Psi) = \gamma\,\Psi^3}. \tag{4}$$

A term proportional to Ψ^3 is the lowest-order odd nonlinearity that preserves the sign of the restoring force; it arises from a quartic potential $V(\Psi) = \frac{1}{4}\gamma\Psi^4$.

Variant B — Density-modulated nonlinearity

Let $\rho(r)$ be the local EMC density inside the soliton and ρ_0 the statutory vacuum background. The deficit fraction is $(1 - \rho(r)/\rho_0)$, and the nonlinear term becomes

$$\boxed{\mathcal{F}(\Psi, \rho) = \gamma\left(1 - \frac{\rho(r)}{\rho_0}\right)\Psi^3}. \tag{5}$$

The nonlinearity is maximal in the core where $\rho(r) \ll \rho_0$ and vanishes in the surrounding vacuum, linking stability directly to the EMC push-out mechanism.

Variant C — Explicit 1-3-6 source-driven dynamics

In the native EWT model the electron consists of ten discrete, mobile wave centres (WCs) arranged in a 1-3-6 configuration. The vector wave equation then takes the driven form

$$\boxed{\left(\frac{\partial^2}{\partial t^2} - c^2\nabla^2\right)\mathbf{\Psi} + \gamma\,\|\mathbf{\Psi}\|^2\,\mathbf{\Psi} = \mathbf{J}_{\text{1-3-6}}(\mathbf{r}, t)}, \tag{6}$$

where the source operator is the sum of ten vectorial centres:

$$\mathbf{J}_{\text{1-3-6}}(\mathbf{r}, t) = \mathbf{j}_{\text{core}}(\mathbf{r} - \mathbf{r}_0) + \sum_{i=1}^{3} \mathbf{j}_{\text{inner}}(\mathbf{r} - \mathbf{r}_i(t)) + \sum_{j=1}^{6} \mathbf{j}_{\text{outer}}(\mathbf{r} - \mathbf{r}_j(t)). \tag{7}$$

Each class carries a distinct phase offset; the phase asymmetry between the inner (3) and outer (6) groups forces a synchronised rotation, generating the 720° spin of the electron.

5 Structural Emergence of the 1-3-6 Geometry

Why the 1-3-6 partition? Three complementary reasons are proposed (pending numerical verification):

1. The **core** (1) anchors the soliton at a single BCC lattice node, providing the central phase reference.
2. The **inner shell** (3) forms an equilateral triangle — the minimal number of vertices that can define a plane whose normal coincides with the spin axis. It is proposed that four or more centres on a single shell at this radius would introduce phase frustration.
3. The **outer shell** (6) is proposed to complete the octahedral coordination of the BCC unit cell, saturating the remaining propagation axes and guaranteeing isotropic far-field emission after spherical masking by the Degraded EMC Wall.

It is proposed that configurations such as 2-3-5 or 2-2-6 would misalign the spin axis or leave BCC axes unmatched, producing a net multipole moment that the nonlinearity cannot cancel. The 1-3-6 triad is therefore the unique self-consistent candidate satisfying: (a) the topological winding number $Q = 10$, (b) the cubic nonlinearity with coefficient γ, and (c) the octahedral symmetry of the $Im\bar{3}m$ BCC lattice. A rigorous proof that this configuration minimises the energy functional remains a task for the M4 simulation.

6 Topological Selection Rule

On the spherical shell S^2 that encloses the soliton, the field configuration is characterised by an integer winding number

$$Q = \frac{1}{4\pi} \int_{S^2} \Psi^* \, \omega \in \mathbb{Z}, \tag{8}$$

where ω is the normalised area form on S^2. For the electron $Q = K_{WC} = 10$.

This number is a topological invariant — it cannot change under continuous deformations of the wave field. The configuration of $Q = 10$ wave centres is a natural candidate for the lowest-energy stable homotopy class that can establish a fully isotropic boundary condition on S^2 (analogous to the global minimum of the spherical Thomson problem). A rigorous proof that $Q = 10$ yields a deeper potential well than $Q = 8, 9, 11$ requires an explicit evaluation of the energy functional with $\gamma\Psi^3$ and the $Im\bar{3}m$ projection — the immediate objective of the M4 simulation.

Equation (1) together with any of the three nonlinear variants and $Q = 10$ constitutes a well-posed, falsifiable model. The stability of the 1-3-6 configuration is at present a postulate, not a proof.

1. The **nonlinearity** (γ) creates a deep, narrow potential well that resists dispersion.
2. The **topological winding number** is proposed to fix the ground state at exactly ten wave centres, pending numerical confirmation that $Q = 10$ is energetically preferred over neighbouring integers.
3. The **BCC stiffness constant** N_{final} (or its geometric limit $8\pi^4$) fixes the coupling without adjustable parameters.

7 Dimensional Weight Operator

Each factor of π in the coupling budget corresponds to a distinct geometric degree of freedom of the soliton. Labelling these as $\pi^{(1)}$ (wave amplitude), $\pi^{(2)}$ (frequency), $\pi^{(3)}, \pi^{(4)}, \pi^{(5)}$ (3D spatial substrate), $\pi^{(6)}$ (soliton radius), and $\pi^{(7)}$ (charge), the coefficient (3) can be factorised as

$$\gamma = \frac{1}{\epsilon_M} = N_{\text{final}}\,\pi^3 \;\approx\; 8\pi^7 = 8 \;\cdot\; \pi^{(1)}\pi^{(2)}\pi^{(3)}\pi^{(4)}\pi^{(5)}\pi^{(6)}\pi^{(7)}. \tag{9}$$

The approximate equality with $8\pi^7$ reflects the geometric limit γ_{geo}; the exact factorisation in terms of labelled $\pi^{(k)}$ is a formal bookkeeping device valid for any numerical prefactor. The factor 8 corresponds to the eight primary propagation axes of the $Im\bar{3}m$ unit cell, ensuring isotropic nonlinear restoring force.

8 Degraded EMC Wall

The nonlinear stabilisation requires a boundary condition: the Degraded EMC Wall, where the EMC density transitions from the deep core deficit to the statutory background ρ_0.

A natural length scale for the wall radius is provided by the electron Compton wavelength λ_C. Using the standard QED relations $r_e = \alpha\lambda_C/2\pi$ and $1/\alpha \approx 137$,

$$\boxed{r_{\text{wall}} \sim \frac{\lambda_C}{2\pi} = \frac{r_e}{\alpha} \approx 137\, r_e}. \tag{10}$$

Because α is itself derivable from BCC geometry, this scale introduces no new parameter. The wall provides a broad transition zone ($\sim 137\, r_e$) within which the nonlinearity can adiabatically relax the EMC density, suppressing sharp gradients that would otherwise destabilise the standing wave.

Possible EMC Density Peak

Two density profiles are compatible with the integral deficit: a monotonic approach to ρ_0, or a local peak ($\rho_{\text{wall}} > \rho_0$) caused by EMC accumulation. If a peak exists, it would assist stabilisation in two ways:

1. In Variant B, the factor $(1-\rho/\rho_0)$ changes sign within the peak, creating a local defocusing barrier that prevents the standing wave from leaking outward.

2. In all variants, a density peak implies a locally higher elastic modulus, acting as a partial reflector that increases the Q-factor of the soliton resonator.

Whether the peak is indispensable or auxiliary can only be decided once the full radial EMC profile is computed from the nonlinear wave equation in the M4 simulation.

9 Internal Wave-Centre Dynamics (Yee Hypothesis)

An additional hypothesis, put forward in the foundational EWT literature, proposes that the electron's ten wave centres are not static but undergo continuous micro-motion around their nodal positions. A displaced centre experiences a restoring force toward the local amplitude minimum; this perpetual shifting is the origin of the 720° spin and, simultaneously, a dynamic stabilisation mechanism. The 1-3-6 geometry ensures that only a fraction of centres are off-node at any instant, while the remainder anchor the structure. This hypothesis should be tested in M4 once the nonlinear term and the EMC Wall are in place: comparing simulations with and without WC motion will reveal whether internal dynamics is essential or secondary.

10 Relation Between Energetic and Topological Arguments

Two complementary conditions single out $K_{WC} = 10$:

1. **Energetic:** The r^5/r^3 scaling disparity creates a deep potential well for intermediate wave-centre counts. K_{WC} cannot be too small (shallow well) or too large (volumetric overload).
2. **Topological:** The winding number Q must be an integer. Only discrete values are permitted.

These are not independent post-hoc justifications; they are mutually reinforcing. The specific value $K = 10$ emerges as the unique integer that simultaneously satisfies both constraints under the octahedral symmetry of the $Im\bar{3}m$ BCC lattice. A rigorous proof that no other integer passes both filters remains a task for the M4 simulation.

11 Summary and Implementation Recommendations

- **Wave equation:** $(\partial_t^2 - c^2\nabla^2)\Psi + \mathcal{F}(\Psi) = 0$.
- **Coupling:** $\gamma = 1/\epsilon_M = N_{\text{final}}\pi^3 \approx 2.41 \times 10^4$. Use γ_{final} initially; the 0.058% difference from $\gamma_{\text{geo}} = 8\pi^7$ may be decisive.
- **Three variants for $\mathcal{F}$:**
  - A. $\gamma\Psi^3$ (pure cubic, minimal implementation).
  - B. $\gamma(1 - \rho/\rho_0)\Psi^3$ (density-modulated, links stability to the EMC push-out deficit).
  - C. $\gamma\|\mathbf{\Psi}\|^2\mathbf{\Psi}$ with an explicit 1-3-6 source operator (directly encodes the electron's wave-centre architecture).
- **Degraded EMC Wall:** $r_{\text{wall}} \sim r_e/\alpha \approx 137\, r_e$. A possible density peak would further assist stabilisation.
- **Topological selection:** $Q = 10$ on S^2 is proposed as the selection rule; rigorous proof that $Q = 10$ is energetically preferred remains an M4 task.
- **1-3-6 partition:** The unique self-consistent candidate; its stability is a postulate pending numerical verification.
- **Energetic + topological arguments:** Complementary constraints that together single out $K_{WC} = 10$.
- **Model status:** Well-posed and falsifiable; stability of the 1-3-6 configuration is a postulate whose numerical verification is the immediate objective of the OpenWave M4 simulation.

Document prepared for the OpenWave development team.

References

[1] Smoliński, Ł. (2025). *The Geometric Identity of Gravity and Dimensional Unification Resolving α, Lepton $(g-2)_l$, Weinberg, and Cabibbo Mixing.* DOI: 10.5281/zenodo.17654657.